香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

COMP 5212

Machine Learning

Instructor: Zihan Zhang

Course website: https://zsubfuncz.github.io/COMP5212-2026Fall/

Canvas code: 72122

Pre-requisite

- Probability
  - Distribution, random variable, expectation, conditional distribution, variance
- Linear algebra
  - Matrix multiplication
- Calculus
- Python programming
  - Numpy, torch

Grading

- 5 assignments (40%)
  - 3 Written assignments (20%) + 2 Programing assignments (20%)
  - 3 free late days in total, for additional late days, 20% penalization applied for each day late
  - No assignment will be accepted more than 3 days late
- Mid-term Exam, open-note (25%)
- Final exam, open-note (35%)

Honor Code

Do's

- Form study groups to discuss (e.g. homework)
- Write down the homework solutions independently
- Write down the names of people with whom you've discussed the homework
- You are encouraged to use generative AI (e.g. ChatGPT) to assist you

Don'ts

- Copy, refer to, or look at solutions from previous years, online, or others
- Copy ChatGPT's answers
- Longer versions on the course website

We have zero tolerance — in the case of honor code violation for a single time, you will fail this course directly

Waiting List & Audit

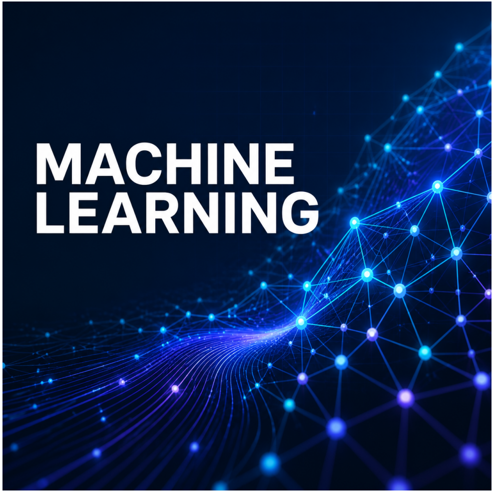
- We will try our best to include more students on the waiting list.
- Audit is allowed, and will be graded in the same criterion

More Info on Course Website

- Canvas is the main platform for announcement, discussion and homework submission
- Recorded videos on canvas
- Syllabus, slides and relevant reading materials
- Detailed course logistics

Overview of Machine Learning

What is Machine Learning



$$\text{Data} \downarrow \boxed{\text{Learning Algorithm}} \downarrow \text{Knowledge}$$

Machine Learning is Trending

- Everywhere — wide application
  - Finance, data scientist, medical diagnosis, translation, self-driving....
- Foundation of artificial intelligence — one of the most important technology for the society in the next 10s of years
  - ChatGPT, large language model, large multimodal model

Foundations of Machine Learning

- Data: where does the datasets come from?
- Candidate set of models: which models are plausible?
- Learning algorithm: how to find the best model to fit the data?

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  - Closed form solutions
  - Optimization methods

Machine Learning Paradigms

- Supervised learning
- Unsupervised learning
- Self-supervised learning
- Reinforcement learning

Supervised Learning

Housing Price Prediction

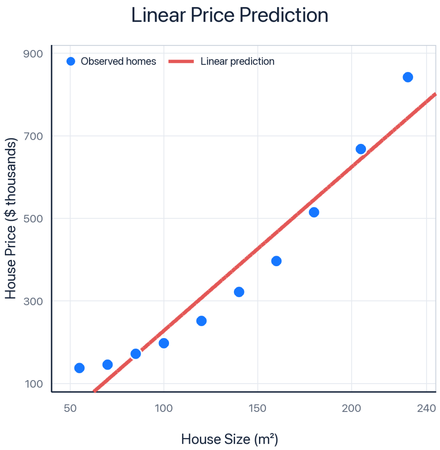
- Given: a dataset that contains n samples

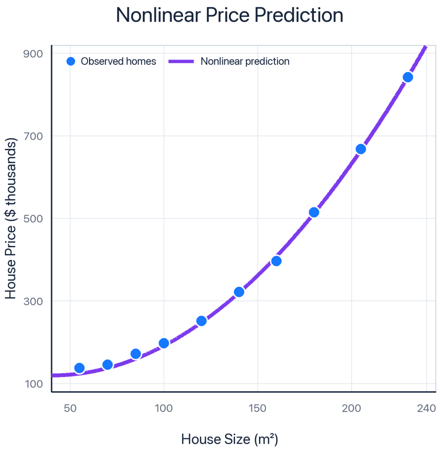
$$\left(x^{(1)}, y^{(1)}\right), \ldots, \left(x^{(n)}, y^{(n)}\right)$$

- Task: if a residence has x square feet, predict its price?

Example from Stanford CS229

Housing Price Prediction





More Features

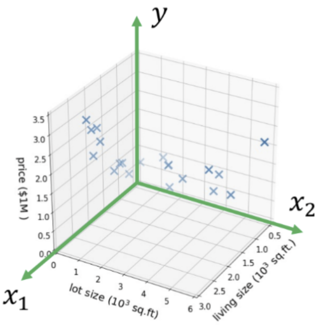
- Suppose we also know the lot size

Example from Stanford CS229

More Features

- Suppose we also know the lot size
- Task: find a function that maps

$$\underbrace{(\text{size},\ \text{lot size})}_{\substack{\text{features/input} \\ x \in \mathbb{R}^2}} \longrightarrow \underbrace{\text{price}}_{\substack{\text{label/output} \\ y \in \mathbb{R}}}$$



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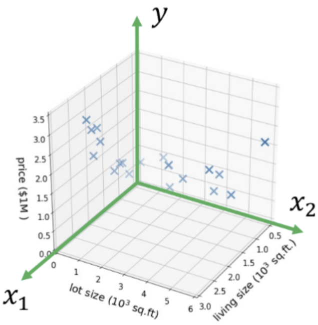
- Dataset: $(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)})$

where $\quad x^{(i)} = \left(x_1^{(i)}, x_2^{(i)}\right)$

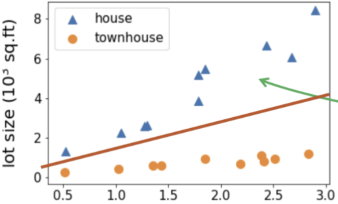


High-dimensional Features

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ \vdots \\ x_d \end{bmatrix} \quad \begin{array}{l} \text{— living size} \\ \text{— lot size} \\ \text{— \# floors} \\ \text{— condition} \\ \text{— zip code} \\ \vdots \end{array} \quad \longrightarrow \quad y \text{ — price}$$

Regression vs Classification

- Regression: if $y \in \mathbb{R}$ is a continuous variable
  - E.g., price prediction
- Classification: the label is a discrete variable
  - E.g., predicting the types of residence

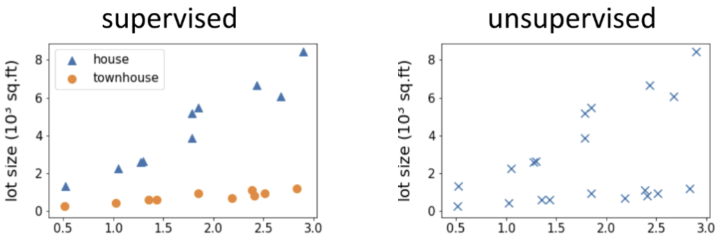


y = house or townhouse?

Unsupervised Learning

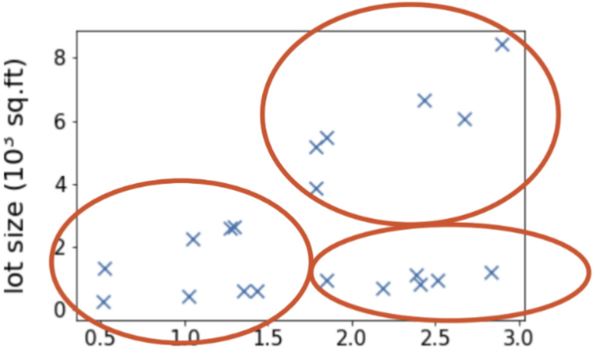
Unsupervised Learning

- The dataset contains no labels: $x^{(1)}, \ldots, x^{(n)}$
- The goal is typically vague: find interesting structures in the data



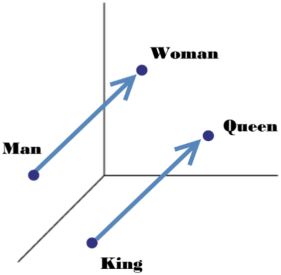
Examples from CS229

Clustering



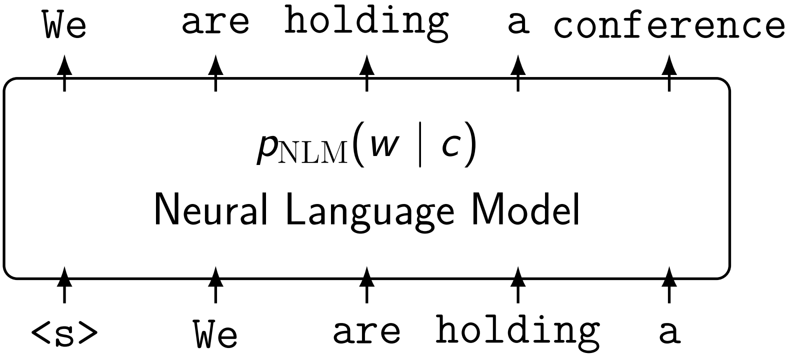
Word Embeddings

WIKIPEDIA
The Free Encyclopedia



Self-supervised Learning

Language Models



The label is from the data itself.

Large Language Models

An autoregressive model predicts each token using only the tokens that appear before it:

$$P(x_1, \ldots, x_T) = \prod_{t=1}^{T} P(x_t \mid x_1, \ldots, x_{t-1}).$$

Training examples

Input context	**Target**
`The`	`cat`
`The cat`	`sat`
`The cat sat`	`on`
`The cat sat on`	`the`
`The cat sat on the`	`mat`

Generation

`Machine`

$\longrightarrow$ `Machine learning`

$\longrightarrow$ `Machine learning can`

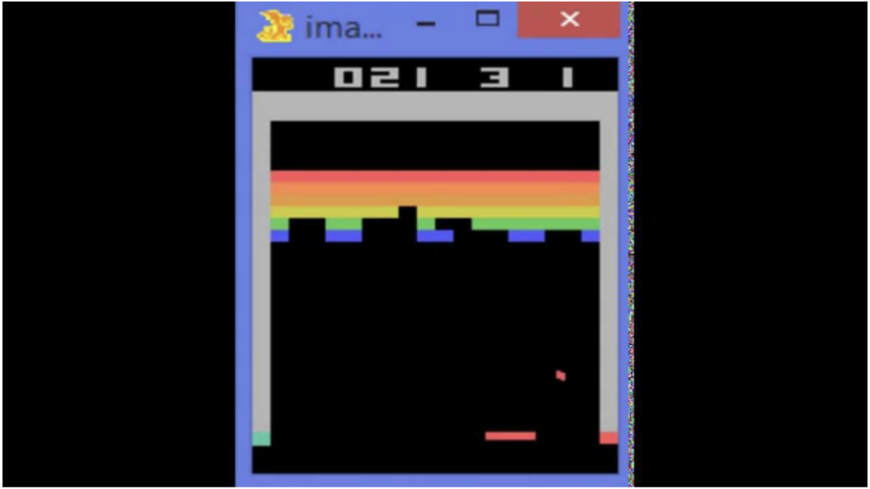
$\longrightarrow$ `Machine learning can predict`

$\longrightarrow \cdots$

Reinforcement Learning

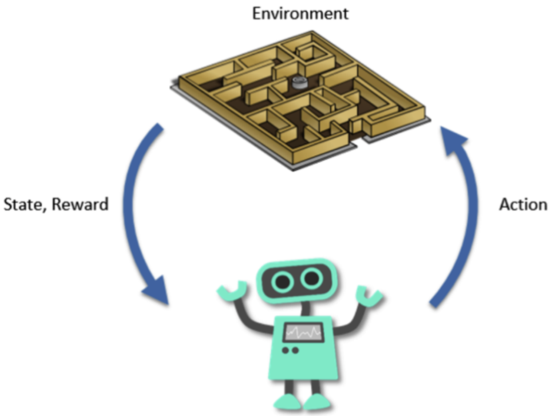
AlphaGo

Atari Breakout Game



Reinforcement Learning

- RL can collect data interactively



Other Learning Paradigms

Overview of Topics

Introduction
Math basics
Foundations of machine learning
Linear/Logistic regression
Generalized linear models, classification
Kernel methods
SVM
MLE
Gradient descent, SGD, Newton's method
Generalization, bias-variance tradeoff
Clustering
Expectation Maximization
PCA/ICA
mid-term exam

Basics of FNN
Neural networks architectures (CNN/RNN)
Variational autoencoder
Generative adversarial networks
Stable diffusion
Reinforcement Learning
Large language models

Thank You!

Questions?